

Challenge (Habanero)

- Q1.** A recipe calls for $3 \cdot 2$ cups of flour to make 6 cupcakes. How many cups are needed for 12 cupcakes?
(A) 6
(B) 8
(C) 12
(D) 18
(E) 24
- Q2.** A construction site has -5 degrees Celsius temperature. If it rises by 8 degrees, what is the new temperature?
(A) -13
(B) -3
(C) 3
(D) 8
(E) 13
- Q3.** Which of the following numbers is an example of an even natural number?
(A) 1
(B) 2
(C) 3
(D) -4
(E) 0
- Q4.** A builder needs to place 15 bricks in a row, with 3 bricks for each section. How many sections can be made?
(A) 3
(B) 4
(C) 5
(D) 6
(E) 7
- Q5.** Which of the following patterns will generate only odd numbers: 1, 3, 5, ...?
(A) Add 1 each time
(B) Add 2 each time
(C) Add 3 each time
(D) Subtract 2 each time
(E) Multiply by 2 each time
- Q6.** If $-2 \cdot 6 = ?$
(A) -8
(B) -12
(C) 12
(D) 8
(E) -18
- Q7.** A recipe for making cookies calls for $\frac{3}{4}$ cup of sugar. If you want to make 2 times the recipe, how much sugar is needed?
(A) $1\frac{1}{2}$ cups
(B) $1\frac{1}{4}$ cups
(C) $1\frac{1}{8}$ cups
(D) $1\frac{3}{4}$ cups
(E) $2\frac{1}{2}$ cups
- Q8.** What is the value of $6 - (-2)$?
(A) 4
(B) 6
(C) 8
(D) 10
(E) 12

Open Ended Questions (FRQ)

- Q9.** A construction crew is building a bridge with an arch that has a height of 5 meters above the water level. The arch is 8 meters wide, and the crew wants to add a decoration that will hang 2 meters below the arch. If the current water level is at -3 meters, what is the distance from the top of the arch to the water level?
- Q10.** A chef has a recipe for 24 cupcakes, and she wants to make 2 batches. If she already has 16 cupcakes made, how many more cupcakes does she need to make to have 2 complete batches?

Chef's Tips

- Allow students to use a calculator for only questions 9 and 10
- Have students show all work and explain their reasoning
- Remind students that answers must be in simplest form